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Combined liquid and air cooled power electronics assembly

Abstract

A power electronics assembly includes one or more power electronics devices, and a heat exchanger to which the one or more power electronics devices are mounted. The heat exchanger includes a heat exchanger body, and one or more fluid pathways extending through the heat exchanger body, the heat exchanger configured to transfer thermal energy from the one or more power electronics devices into a flow of fluid passing through the one or more fluid pathways. A plurality of fins extend from an outer surface of the heat exchanger body. The flow of fluid is a flow of liquid refrigerant diverted from a condenser of a heating, ventilation, and air conditioning (HVAC) system.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/274,762 filed Nov. 2, 2021, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(1) Exemplary embodiments pertain to the art of heat exchangers, and more particularly to heat

exchangers for cooling power electronics.

(2) Power electronics devices such as motor drives generate waste heat during operation of the device. Additionally, when the power electronics devices heat up the operational efficiency of the devices can degrade adding to the amount of heat generated. When utilized in a refrigeration system to drive, for example, a compressor of the refrigeration system, effective thermal integration of these devices can be an important aspect to the system's overall efficiency and reliability.

Consequently, a goal of the system integrator is to maintain these components within a range of operating temperatures which will maximize the system efficiency. One means of integrated cooling is to provide refrigerant from the refrigeration system to cool the power electronics. However, when you startup or shutdown the refrigeration system there may not be enough liquid circulating to cool the power electronics, which may lead to overheating during startup/shutdown. One way to mitigate this issue is to provide a dedicated pump to ensure liquid is available during these times, but this is an expensive solution. Accordingly, there remains a need in the art for a more effective solution to provide integrated cooling for the power electronics that ensures, or at least mitigates, overheating during startup/shutdown.

BRIEF DESCRIPTION

(3) In one embodiment, a power electronics assembly includes one or more power electronics devices, and a heat exchanger to which the one or more power electronics devices are mounted. The heat exchanger includes a heat exchanger body, and one or more fluid pathways extending through the heat exchanger body, the heat exchanger configured to transfer thermal energy from the one or more power electronics devices into a flow of fluid passing through the one or more fluid pathways. A plurality of fins extend from an outer surface of the heat exchanger body. The flow of fluid is a flow of liquid refrigerant diverted from a condenser of a heating, ventilation, and air conditioning (HVAC) system.

(4) Additionally or alternatively, in this or other embodiments the one or more power electronics devices are secured to a first outer surface of the heat exchanger body, and the plurality of fins extend from a second outer surface of the heat exchanger body different from the first outer surface.

(5) Additionally or alternatively, in this or other embodiments the second outer surface is opposite the first outer surface.

(6) Additionally or alternatively, in this or other embodiments the heat exchanger body is formed from two or more plates secured at a mating surface.

(7) Additionally or alternatively, in this or other embodiments each plate includes a portion of the one or more fluid pathways.

(8) Additionally or alternatively, in this or other embodiments the one or more power electronics devices includes a variable frequency drive of a heating, ventilation, and air conditioning (HVAC) system.

(9) In another embodiment, a power electronics assembly includes one or more power electronics devices, and a housing including two or more housing walls to partially enclose the one or more power electronics devices. A housing wall of the two or more housing walls is configured as a heat exchanger. The heat exchanger includes a heat exchanger body and one or more fluid pathways extending through the heat exchanger body, the heat exchanger configured to transfer thermal energy from the one or more power electronics devices into a flow of fluid passing through the one or more fluid pathways. A plurality of fins extend from an outer surface of the heat exchanger body. The flow of fluid is a flow of liquid refrigerant diverted from a condenser of a heating, ventilation, and air conditioning (HVAC) system.

(10) Additionally or alternatively, in this or other embodiments the one or more power electronics devices are secured to a first outer surface of the heat exchanger body, and the plurality of fins extend from a second outer surface of the heat exchanger body different from the first outer surface.

- (11) Additionally or alternatively, in this or other embodiments the second outer surface defines an exterior surface of the housing.
- (12) Additionally or alternatively, in this or other embodiments the second outer surface is opposite the first outer surface.
- (13) Additionally or alternatively, in this or other embodiments the heat exchanger body is formed from two or more plates secured at a mating surface.
- (14) Additionally or alternatively, in this or other embodiments each plate includes a portion of the one or more fluid pathways.
- (15) Additionally or alternatively, in this or other embodiments the one or more power electronics devices includes a variable frequency drive of a heating, ventilation, and air conditioning (HVAC) system.
- (16) In yet another embodiment, a method of cooling one or more power electronics devices includes securing the one or more power electronics devices to a heat exchanger, the heat exchanger including one or more fluid pathways and a plurality of fins. Circulating a flow of fluid through one or more fluid pathways to transfer thermal energy from the one or more power electronics to the flow of fluid passing through the one or more fluid pathways. The thermal energy is dissipated from the heat exchanger via the plurality of fins extending from an outer surface of the heat exchanger. The flow of fluid is a flow of liquid refrigerant diverted from a condenser of a heating, ventilation, and air conditioning (HVAC) system.
- (17) Additionally or alternatively, in this or other embodiments the one or more power electronics devices are secured to a first outer surface of the heat exchanger body, and the plurality of fins extend from a second outer surface of the heat exchanger body different from the first outer surface.
- (18) Additionally or alternatively, in this or other embodiments the second outer surface is opposite the first outer surface.
- (19) Additionally or alternatively, in this or other embodiments the heat exchanger body is formed from two or more plates secured at a mating surface.
- (20) Additionally or alternatively, in this or other embodiments each plate includes a portion of the one or more fluid pathways.
- (21) Additionally or alternatively, in this or other embodiments the one or more power electronics devices includes a variable frequency drive of a heating, ventilation, and air conditioning (HVAC) system.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
- (2) FIG. 1 is an illustration of an embodiment of a power electronics assembly;
- (3) FIG. 2 is an illustration of an embodiment of a heating, ventilation, and air conditioning (HVAC) system;
- (4) FIG. 3 is an illustration of an embodiment of a heat exchanger of a power electronics assembly;
- (5) FIG. 4 is a cross-sectional view of an embodiment of a heat exchanger; and
- (6) FIG. 5 is a perspective view of an embodiment of an enclosed power electronics assembly.

DETAILED DESCRIPTION

- (7) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.
- (8) Illustrated in FIG. 1 is an embodiment of a power electronics assembly **10**. The power electronics assembly **10** includes one or more power electronics devices **12** mounted to a heat

exchanger **14** utilized to reject and dissipate thermal energy generated by the power electronics devices **12** during operation. In some embodiments, the power electronics devices **12** include a variable frequency drive (VFD) operably connected to a compressor **16** of a heating, ventilation, and air conditioning (HVAC) system **18**.

(9) Referring to FIG. 2, the HVAC system **18** includes, for example, a vapor compression circuit **70** including the compressor **16**, a condenser **72**, an expansion device **74** and an evaporator **76** arranged in series and having a volume of refrigerant flowing therethrough. The power electronics assembly **10** is connected to the vapor compression circuit **70** such that a portion of the flow of liquid refrigerant exiting the condenser **72** is diverted through the heat exchanger **14**, bypassing the expansion device **74** and is directed from the heat exchanger **14** to the evaporator **76**. The liquid refrigerant from the condenser **72** absorbs thermal energy generated by the power electronics devices **12**. This thermal energy generated by the power electronics devices **12** may cause the refrigerant passing through the heat exchanger **14** to boil, changing phase of the refrigerant from liquid to vapor inside the heat exchanger **14**.

(10) Referring now to FIG. 3, an exemplary heat exchanger **14** is illustrated in more detail. As shown, the heat exchanger **14** includes a plurality of enclosed fluid pathways **20** extending therethrough, enabling a flow of fluid **22** such as, for example, refrigerant, glycol, or water to flow therein. The fluid pathways **20** extend between a heat exchanger inlet **26** through which the flow of fluid **22** enters the heat exchanger **14**, and a heat exchanger outlet **24** through which the flow of fluid **22** exits the heat exchanger **14**. The heat exchanger inlet **26** is connected to the fluid pathways **20** via an inlet manifold **30** to distribute the flow of fluid **22** to the fluid pathways **20**, and similarly the heat exchanger outlet **24** is connected to the fluid pathways **20** via an outlet manifold **28**. Immediately downstream of the inlet manifold **30**, orifices **80** of different diameters, increasing with increasing distance from the heat exchanger inlet **26** are provided to distribute the flow of fluid **22** from the inlet manifold **30** equally among all of the fluid pathways **20**. The enhancement features described in the following text will be in the fluid pathways **20** downstream of the respective orifices **80**.

(11) In some embodiments, the heat exchanger **14** is formed from a metal material, such as aluminum, aluminum alloy, steel, steel alloy, copper, copper alloy, or the like, and referring again to FIG. 1, may be formed from two or more plates **32**, **34** abutting one another along a side and joined using any suitable means such as brazing, welding, clamping, compressing, bolting, and the like. The plates **32**, **34** may each include a portion of the fluid pathways **20**, the inlet manifold **30**, the outlet manifold **28**, the heat exchanger inlet **26** and/or the heat exchanger outlet **24** formed therein. The mating surfaces of the plates **32**, **34** can be configured to correspond to one another, e.g., to fit together and seal the fluid circuit therebetween. The mating surfaces of the plates **32**, **34** can include precision surfaces formed from a process having highly accurate and precise dimensional control, such as through computer numerical control (CNC) machining process and/or net shape, or near net shape manufacturing process. Optionally or additionally, a sealing material can be disposed between the plates **32**, **34** to aide in preventing leakage from the fluid circuit.

(12) In operation, the flow of fluid **22**, liquid refrigerant from the condenser **72**, enters the heat exchanger **14** at the heat exchanger inlet **26** and is distributed to the fluid pathways **20** via the inlet manifold **30**. The heat exchanger **14** conducts thermal energy from the power electronics devices **12** and thermal energy is exchanged with the flow of fluid **22** flowing through the fluid pathways **20**, resulting in cooling of the power electronics devices **12**. The vapor flow of fluid **22** is then collected at the outlet manifold **28** and exits the heat exchanger **14** at the heat exchanger outlet **24**.

(13) In the present disclosure, features of the heat exchanger **14** are disclosed which improve performance of the heat exchanger **14**, many of which are features to improve boiling performance of the flow of fluid **22**, the refrigerant from the condenser **72**. The flow of fluid **22** provided to the heat exchanger **14** may be diverted from the condenser **72** and in some embodiments enters the heat exchanger **14** in a liquid phase with 0% vapor quality or up to 20% vapor quality. The presently

disclosed heat exchanger **14** may be operated to ensure the vapor quality of the flow of fluid **22** exiting the heat exchanger **14** has a vapor quality of from about 25% to about 80%, or in another embodiment from about 40% to about 60%, or in yet another embodiment from about 45% to about 55%, or about 50%. The vapor quality change happens as the heat from power electronics devices **12** is absorbed by the flow and used for phase change of the liquid refrigerant to vapor while travelling from the heat exchanger inlet **26** to the heat exchanger outlet **24**.

(14) As shown in FIGS. **4** and **5**, the power electronics devices **12** may be secured to a first lateral side **36** of the heat exchanger **14**, and a plurality of fins **38** may extend from a second lateral side **40** of the heat exchanger **14**, opposite the first lateral side **36**. The plurality of fins **38** extend outwardly from the second lateral side **40** and aid in dissipating heat from the power electronics devices **12** and the heat exchanger **14** via airflow passing through the plurality of fins **38**.

(15) The power electronics devices **12** are packaged in a housing **42** that has a plurality of housing walls **44** at least partially enclosing the power electronics devices **12**. At least one housing wall **44** may be configured as a heat exchanger **14**, having the fluid pathways **20** embedded in the housing wall **44** and the plurality of fins **38** extending outwardly from an exterior housing wall surface **46**. The heat exchangers **14** serving as the housing walls **44** may allow the heat exchanger **14** to serve as both a structural element for mounting the power electronics devices **12**, and as components of the housing **42**. The configuration may ensure that the second lateral side **40** and the plurality of fins **38** are exposed to ambient air and rejects heat via natural convection when the flow of fluid **22** through the plurality of fluid pathways **20** is not established, such as during system startup or shutdown. Further, in configurations where more than one housing wall **44** is configured as a heat exchanger **14** surface area for heat rejection may greatly increase.

(16) By configuring one or more housing walls **44** as heat exchangers **14**, cost of the power electronics assembly **10** may be reduced by eliminating the separator housing walls, and volume and performance of the assembly **10** is improved due to improved cooling performance and boiling of the flow of fluid **22** in the heat exchanger **14**.

(17) The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application.

(18) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(19) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A power electronics assembly, comprising: one or more power electronics devices; and a heat exchanger to which the one or more power electronics devices are mounted, the heat exchanger comprising: a heat exchanger body; one or more fluid pathways extending through the heat exchanger body, the heat exchanger configured to transfer thermal energy from the one or more

power electronics devices into a flow of fluid passing through the one or more fluid pathways; and a plurality of fins extending from an outer surface of the heat exchanger body; wherein the flow of fluid is a flow of liquid refrigerant diverted from a condenser of a heating, ventilation, and air conditioning (HVAC) system, bypassing an expansion device of the HVAC system; wherein a heat exchanger inlet of the heat exchanger is fluidly connected to the HVAC system between the condenser and the expansion device; and wherein a heat exchanger outlet of the heat exchanger is fluidly connected to the HVAC system between the expansion device and an evaporator of the HVAC system.

2. The power electronics assembly of claim 1, wherein the one or more power electronics devices are secured to a first outer surface of the heat exchanger body, and the plurality of fins extend from a second outer surface of the heat exchanger body different from the first outer surface.
3. The power electronics assembly of claim 2, wherein the second outer surface is opposite the first outer surface.
4. The power electronics assembly of claim 1, wherein the heat exchanger body is formed from two or more plates secured at a mating surface.
5. The power electronics assembly of claim 4, wherein each plate comprises a portion of the one or more fluid pathways.
6. The power electronics assembly of claim 1, wherein the one or more power electronics devices comprises a variable frequency drive of the HVAC system.
7. A power electronics assembly, comprising: one or more power electronics devices; a housing comprising two or more housing walls to partially enclose the one or more power electronics devices; wherein a housing wall of the two or more housing walls is configured as a heat exchanger, the heat exchanger comprising: a heat exchanger body; and one or more fluid pathways extending through the heat exchanger body, the heat exchanger configured to transfer thermal energy from the one or more power electronics devices into a flow of fluid passing through the one or more fluid pathways; and a plurality of fins extending from an outer surface of the heat exchanger body; wherein the flow of fluid is a flow of liquid refrigerant diverted from a condenser of a heating, ventilation, and air conditioning (HVAC) system, bypassing an expansion device of the HVAC system; wherein a heat exchanger inlet of the heat exchanger is fluidly connected to the HVAC system between the condenser and the expansion device; and wherein a heat exchanger outlet of the heat exchanger is fluidly connected to the HVAC system between the expansion device and an evaporator of the HVAC system.
8. The power electronics assembly of claim 7, wherein the one or more power electronics devices are secured to a first outer surface of the heat exchanger body, and the plurality of fins extend from a second outer surface of the heat exchanger body different from the first outer surface.
9. The power electronics assembly of claim 7, wherein the second outer surface defines an exterior surface of the housing.
10. The power electronics assembly of claim 8, wherein the second outer surface is opposite the first outer surface.
11. The power electronics assembly of claim 7, wherein the heat exchanger body is formed from two or more plates secured at a mating surface.
12. The power electronics assembly of claim 11, wherein each plate comprises a portion of the one or more fluid pathways.
13. The power electronics assembly of claim 7, wherein the one or more power electronics devices comprises a variable frequency drive of the HVAC system.
14. A method of cooling one or more power electronics devices, comprising: securing the one or more power electronics devices to a heat exchanger, the heat exchanger comprising one or more fluid pathways and a plurality of fins; circulating a flow of fluid through one or more fluid pathways to transfer thermal energy from the one or more power electronics to the flow of fluid passing through the one or more fluid pathways; and dissipating the thermal energy from the heat

exchanger via the plurality of fins extending from an outer surface of the heat exchanger; wherein the flow of fluid is a flow of liquid refrigerant diverted from a condenser of a heating, ventilation, and air conditioning (HVAC) system, bypassing an expansion device of the HVAC system; wherein a heat exchanger inlet of the heat exchanger is fluidly connected to the HVAC system between the condenser and the expansion device; and wherein a heat exchanger outlet of the heat exchanger is fluidly connected to the HVAC system between the expansion device and an evaporator of the HVAC system.

15. The method of claim 14, wherein the one or more power electronics devices are secured to a first outer surface of the heat exchanger body, and the plurality of fins extend from a second outer surface of the heat exchanger body different from the first outer surface.

16. The method of claim 15, wherein the second outer surface is opposite the first outer surface.

17. The method of claim 14, further comprising forming the heat exchanger body from two or more plates secured at a mating surface.

18. The method of claim 17, wherein each plate comprises a portion of the one or more fluid pathways.

19. The method of claim 14, wherein the one or more power electronics devices comprises a variable frequency drive of the HVAC system.

20. The power electronics assembly of claim 1, wherein: the one or more fluid pathways is two or more fluid pathways; the heat exchanger includes an inlet manifold connecting the heat exchanger inlet to the two or more fluid pathways; wherein a plurality of orifices connect the inlet manifold to the two or more fluid pathways, the plurality of orifices having an increasing orifice diameter with increasing distance from the heat exchanger inlet.
